INCH-POUND

MIL-F-48877 (AR)
AMENDMENT 9
20 FEBRUARY 1991
SUPERSEDING
AMENDMENT 8
2 MARCH 1990

MILITARY SPECIFICATION

FUZE, ROCKET: M439 PARTS AND ASSEMBLY WITH DETONATOR

This Amendment forms a part of Military Specification MIL-F-48877 (AR), dated 17 September 1982,and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 2.1.2, Delete "9270749 - Box, Fiberboard (Inner) Packing, Ammunition, for Fuze Rocket M439"

* Delete "9270750 - Box, Fiberboard (Outer) Alternative Packing, Ammunition, for Fuze Rocket M439"

* Delete "9270751 - Box, Packing, Ammunition, for Fuze Rocket M439"

Add the following after DRAWINGS:

"9324106 - Rocket management System, Inventory - Deployment, M138

* 9357776 - Packing and Marking of Outer Fiberboard Box for Fuze, Rocket M439".

PAGE 4

3.9, Delete in its entirety.

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PAGE 5

3.14.4.3 Delete "+160° ± 5° F" and substitute "+150° ± 5°F".

3.15.1, Delete in its entirety.

PAGE 6

3.17, Delete in its entirety.

3.18, Delete the first sentence and substitute:

"The safe position of S&A's shall be verified just prior to issuance for fuze final assembly; limited to a maximum of half-a-day's production."

3.19, Delete in its entirety and substitute:

"3.19 Charge circuit continuity (fuze unarmed). The fuze unarmed shall be tested for continuity through its umbilical lead. The contractor shall design a charge circuit continuity test set that provides an output pulse of 40 mamps maximum for a period of 0.10 msec. maximum. The accept/reject requirement of the umbilical cable and the charge circuit of the firing capacitor is:

a. Reject as an open circuit if measured resistance is greater than the resistance of a 2.25 kohm resistance connected in series with the charge circuit of the firing capacitor.

b. Reject as a short if the measured resistance is less than the resistance of a 220 ohm resistance connected in parallel with the charge circuit of the firing capacitor.

The test set shall be calibrated with acceptable production modules using the above modifications. Thirty fuzes shall be modified for the short condition and thirty fuzes shall be modified for the open condition. The results of the calibration process will be subject to government review and approval."

PAGE 7

3.22, Delete "-Non-Arm and Arming Times only".

3.23.3, Delete "+160° ± 5°F" and substitute "+150° ± 5°F"

3.26, Add immediately after "Ballistic firing.":

"The fuze shall perform properly following temperature conditioning for a minimum of 8 hours after stabilization at: -50° ± 5°F, 70° ± 15°F (ambient) and +150° ± 5°F."

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PAGE 8

3.28.1, Add "all required markings shall be neat and legible."

4.3.1, Under "Quantity" delete "810" and substitute "486".

PAGE 10

4.3.1, Delete:

"Pin, Interfacing	9260697	40"

And substitute:

"Pin, Interfacing	9260697-1	10
Pin, Interfacing	9260697-2	10"

PAGE 12

TABLE II: For "Rotor Housing" delete "4.4.2.16" and substitute "4.4.2.6".

PAGE 18

TABLE II: Delete "Module Turn on Time" in its entirety, 2 places. Under "No. of Sample Units" delete "810" and substitute "486".

Sheet 8 of 10, After Thermal Shock, No. OF SAMPLE UNITS, change "60" to "36".

PAGE 19

TABLE II: Electronics Housing Assembly: delete

"Corrosion Resistance of Protective Coating	25	3.4" and
"Umbilical Assembly Pull Test	25	3.4".

PAGE 24

4.4.2.3, Major 103: Delete "Width of detonator groove" and substitute "Location of detonator end."

PAGE 27

4.4.2.6, Category 103: Delete "positin" and substitute "position".

PAGE 29

4.4.2.8, Delete Major 101: "Radius of bend (70° - 2°)" and substitute "Radius of bend (70° + 5°)."

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PAGE 30

4.4.2.9, Delete Major 101, "Angle of bend ($83^{\circ}+4^{\circ}$)" and substitute "Angle of bend ($82^{\circ} + 6^{\circ}$)".

Delete Major 102: "Angle of bend ($62^{\circ} + 5^{\circ}$)" and substitute "Angle of bend ($61^{\circ} + 6^{\circ}$)".

Add Major 107:
"Angle of bend ($64^{\circ} + 6^{\circ}$) 0.40% 3.4 Gage."

PAGE 31

4.4.2.10, Delete Major 102, "Angle of bend ($89^{\circ} + 3^{\circ}$)" and substitute "Angle of bend ($86^{\circ} + 6^{\circ}$)".

Add Major 105:
"Angle of bend ($64^{\circ} + 6^{\circ}$) 0.40% 3.4 Gage."

PAGE 32

4.4.2.11, Delete Major 101, "Angle of bend ($49^{\circ} + 2^{\circ}$)" and substitute "Angle of bend ($45^{\circ} + 6^{\circ}$)".

Delete Major 102, Angle of bend ($86^{\circ} + 3^{\circ}$) and substitute "Angle of bend ($86^{\circ} + 6^{\circ}$)".

Add Major 106:
"Angle of bend ($64^{\circ} + 6^{\circ}$) 0.40% 3.4 Gage".

PAGE 33

4.4.2.12, Delete Major 101, "Angle of bend ($90^{\circ} + 2^{\circ}$)" and substitute "Angle of bend ($90^{\circ} + 4^{\circ}$).

Category 102: delete "positin" and substitute "position".

Add Major 106
"Angle of bend ($64^{\circ} + 6^{\circ}$) 0.40% 3.4 Gage" 2 places.

PAGE 34

4.4.2.13, Delete Major 104: "Angle of bend ($89^{\circ} + 3^{\circ}$) and substitute "Angle of bend ($87^{\circ} + 6^{\circ}$)".

Add Major 107:

"Angle of bend ($64^{\circ} + 6^{\circ}$) 0.40% 3.4 Gage"

PAGE 35

4.4.2.14, Category 109: Delete in its entirety.

Category 210: Delete "places".

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PAGE 36

4.4.2.15, Category 105: Delete "eyelet holes, 5 places" and substitute "dimples", 6 places".

PAGE 43

4.4.2.22, Category 102: Delete in its entirety.

PAGE 45

* 4.4.2.24, Category 103: Delete "Overall height of insulator from side of weight" and substitute "Maximum gap between insulation and side of "G" weight".

Delete Major 104 in its entirety.

Categories 105 and 106: Delete "(a)" and substitute "a,b".

Add note to note block as follows:

"(b) Special Inspection Level S-4 of MIL-STD-105D".

PAGE 46

4.4.2.25, Delete Major 106 in its entirety.

PAGE 47

* 4.4.2.26, Delete Critical 3 in its entirety.

PAGE 48

4.4.2.26, Delete Major 110 in its entirety.

Add Majors 114 and 115 as follows:

"114	Wrong lubricant		100%	3.4	Certification
115	Excessive lubricant	(a)	0.40%	3.4	Visual"

Add note to notes block as follows:

"(a) Excessive lubricant is defined as evidence of non-uniform coating or collection of fluid in visible areas."

PAGE 49

* 4.4.2.27, Add Category 2 as follows:

"Rotor reset capability	---	100%	3.8	4.5.10".

Delete Major 101 in its entirety.

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PAGE 50

4.4.2.27, Category 109: Add "or less than two speed clip retaining detonator".

PAGE 51

4.4.2.25, Delete "4.4.2.25" and substitute, "4.4.2.27".

4.4.2.27, Delete Major 110 and substitute:

"110	Rotor ground strap not contacting detonator socket terminal	0.40%	3.4	Visual".

* Delete Major 111 in its entirety.

Add Major Category 116:

"116	Eyelets too close to surface of bottom plate assembly 2 places	0.40%	3.4	Gage".

PAGE 52

4.4.2.28, Category 201: Delete "3.28" and substitute "3.28/3.4"

PAGE 55

4.4.2.31, Categories 105, 107, 108, 112, and 113: Delete in their entirety.

Category 110: Delete "8 seconds" and substitute "10 seconds".

Category 111: Delete "12 seconds" and substitute "24 seconds".

PAGE 56

4.4.2.31, Delete Minor 203 in its entirety.

Delete Category 204 in its entirety and substitute:.

"204	Excessive molding defects or a component or substrate is exposed	0.65%	3.4	Visual".

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PAGE 57

4.4.2.32, Category 103: Delete "Termination radius of switch assy, connector pin" and "visual" and substitute "Interface tip flat over max." and "gage", respectively.

PAGE 58

4.4.2.33, Category 103: Delete "Radius termination of interface" and substitute "Interface tip flat over max".

Category 104: Delete "4.5.2" and substitute "Gage".

Delete Minor 201 in its entirety.

PAGE 59

4.4.2.34, Categories 102, 103, 106, and 107: Add "(a)" to "EXAMINATION".

Add Major Categories 108, 109, and 110:

"108	True Position of blind hole	0.40%	3.4	Gage
109	Diameter of blind hole	0.40%	3.4	Gage
110	Depth of blind hole, mim	0.40%	3.4	Gage".

Minor 201: Delete "3.28" and substitute "3.28/3.4".

Add note to notes block:

"(a) If pins are molded into housing, categories 106 and 107 do not apply; categories 102 and 103 refer to pins instead of pin holes."

PAGE 60

4.4.2.35, Categories 102 and 103: Delete in their entirety.

PAGE 61

4.4.2.36, Critical 1: Delete "3.17" and substitute "3.18".

PAGE 62

4.4.2.36, Fuze, Rocket, M439: Add Major 109

"109	Module time delay (after centrifuge tear down)	32	3.15.2	4.5.18.1".

PAGE 63

4.4.4.1.2b, Delete "4.5.17".

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PAGE 64

4.4.4.2, Delete "Paragraph 4.5.17 - Module Turn ON Time".

PAGE 65

4.5.2, Add Note 2: "Mold samples may be returned to the lot after inspection."

PAGE 66

4.5.6.1, Delete "Apply an" and substitute "Apply the required".

PAGE 67

4.5.6.2, 4.5.6.3, 4.5.6.4, and 4.5.6.5: Delete "Apply an" and substitute "Apply the required".

4.5.6.6, Add "and removed from the Lot".

4.5.6.7, Delete "outside face of the annular gear" and substitute "end of the stud".

PAGE 69

* 4.5.10, In first sentence delete "loaded".

Add "This test may be run with the S&A loaded or unloaded."

PAGE 70

4.5.1.5, Delete "4.5.1.5" and substitute "4.5.15".

4.5.1.4, Delete "+160° $\pm$ 5°F" and substitute ,150° $\pm$ 5°F".

4.5.17, Delete in its entirety.

4.5.18, Delete "four seconds" and substitute "seven seconds" and after "seven seconds" add "minimum".

4.5.18.1, Add the following paragraph:

"4.5.18.1, Module time delay function test. (After tear down from centrifuge lot Acceptence.) The module shall function within the requirements of Table VI (+70°F only) If two or more do not meet the requirements, the lot shall be rejected".

PAGE 71

4.5.19, Module thermal shock. Last sentence, delete "one defect" and substitute "two defects".

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4.5.20, In last sentence before 4.4.4.2 add "3.18".

PAGE 72

4.5.23, Delete "the nominal 24 second time before" and substitute "20.275 seconds prior to".

PAGE 74

4.5.29, Delete paragraph under Table IV in its entirety.

PAGE 75

4.5.31, Delete entire note at top of page.

4.5.33, Delete in its entirety and substitute:

"4.5.33 Test equipment failure. If the test equipment fails to produce an arming readout (i.e., recording paper skewes, audio pick-up, or amplifier inoperative) during the arming time test as verified by the Government, the Safety and Arming Device shall be visually inspected for arming; if found to be armed or partially armed, score as a "no test" and replace with a spare fuze. The timing record produced by the fuze on the approved fuze timing equipment shall be considered the timing record of the fuze under the applicable defect classification."

4.5.34, Delete Paragraph and substitute:

"4.5.34 Rounding off of fuze function times, fuze arming times and module function times. Fuze function times, fuze arming times, and module function times if recorded to more than two (2) places to the right of the decimal point shall be "rounded" off to the hundredth of a second."

PAGE 76

4.5.36, In the third sentence, delete and substitute "MK66 rocket motors shall be used for all tests."

In the seventh sentence after "ballistic test", add "the fuze shall be removed from the test vehicle and checked for continuity to determine fuze or test vehicle failure. If it is determined that there is a test vehicle failure then it is considered a no-test."

PAGE 77

4.5.36.2, Acceptance criteria.

In box under Nominal Time Setting (sec.):

Delete "2" and substitute "1.0 km (1.975)".

Delete "8" and substitute "3.2 km (8.075)".

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Delete "14".

Delete "20" and substitite "6.0 km (20.275)".

In box under Function time Delay (Sec.):

Delete "7.84 to 2.13" and substitute "1.85 to 2.09".

Delete "7.84 to 8.16" and substitute "7.87 to 8.27".

Delete "13.72 to 14.28".

Delete "19.60 to 20.40" and substitute "19.77 to 20.78".

4.5.36.3.1, First article sample.

Delete "three (3)" and substitute "four (4)".

Add "Also, if three (3) or more out of the four units are duds, the sample shall be rejected. Duds are failed units other than those that do not meet the function time delays or fail the charge circuit continuity test."

4.5.36.3.2, Lot acceptance sample.

Delete "four (4)" and substitute "five (5)".

Add "Also, if three (3) or more out of the fire units are duds, the lot shall be rejected. Duds are failed units other than those that do not meet the function time delays, or fail the charge circuits continuity test."

PAGE 78

* 5.2, Delete in its entirety and substitute:

"5.2 Packing and marking. The packaged fuzes shall be packed and marked as specified on Dwg. 9357776. Or, as alternate, packing and marking shall comply with CFR49, Parts 100- 199."

PAGE 79

6.3.1, Delete "ARRADCOM, ATTN: DRDAR-QAF-T, Dover, NJ 07801" and substitute "Armament Research, Development and Engineering Center, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAF-A(D), Picatinny Arsenal, NJ 07806-5000".

6.5, Delete "ARRADCOM, ATTN: DRDAR-QAF-I" and substitute "Armament Research, Development and Engineering Center, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAF-I(D)".

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PAGE 81

TABLE V: Delete in its entirety and substitute included PAGE (81).

PAGE 82

TABLE VI: Delete in its entirety and substitute included PAGE (82).

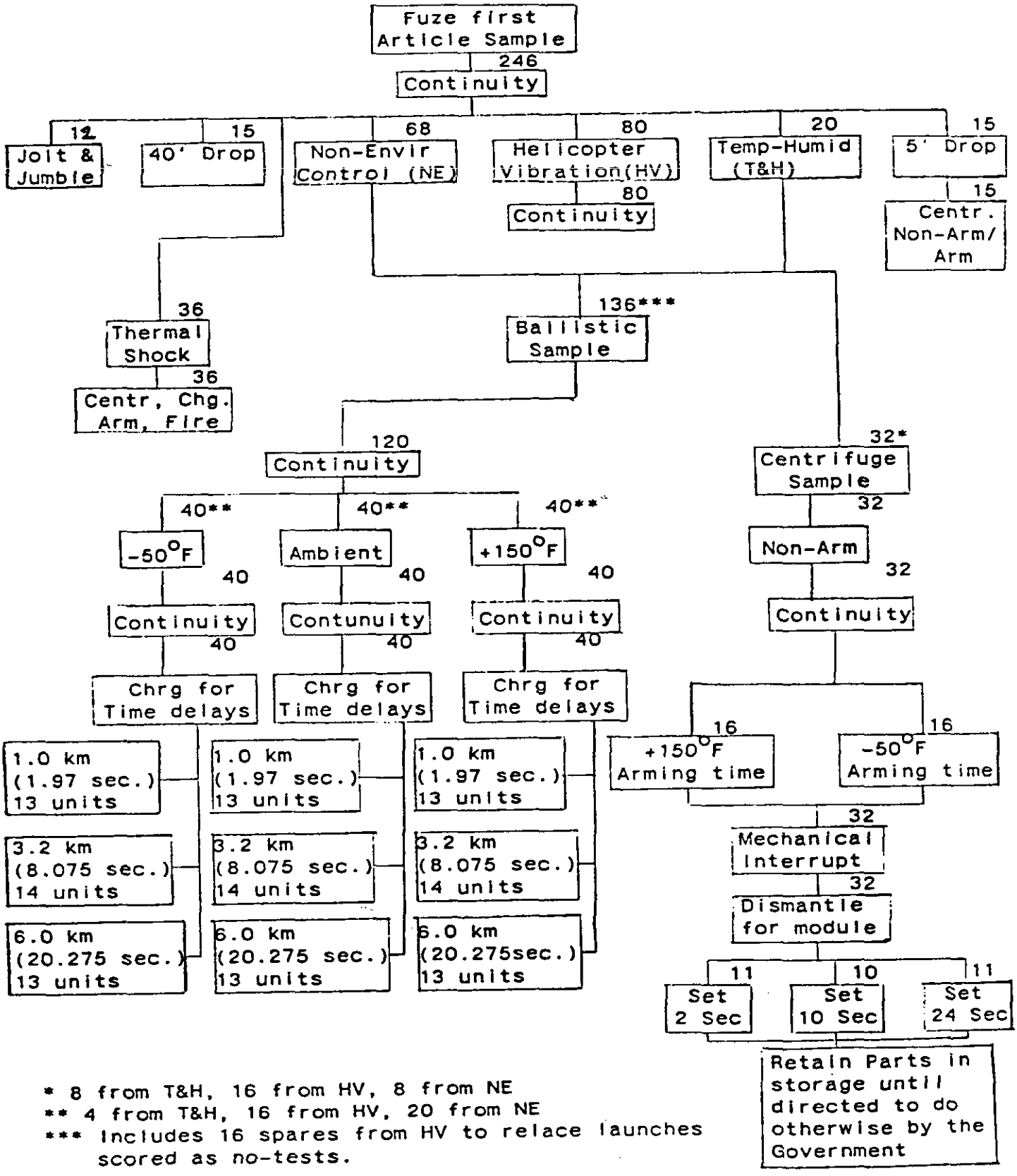
PAGE 83

TABLE VI: Paragraph 3: Delete in its entirety.

PAGE 84

TABLE VII: Delete in its entirety and substitute included PAGE (84).

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* 8 from T&H, 16 from HV, 8 from NE
** 4 from T&H, 16 from HV, 20 from NE
*** Includes 16 spares from HV to relace launches scored as no-tests.

Note: A setter that represents the helicopter setter system shall be used for delay time acceptance tests in all cases.

TABLE V First article test

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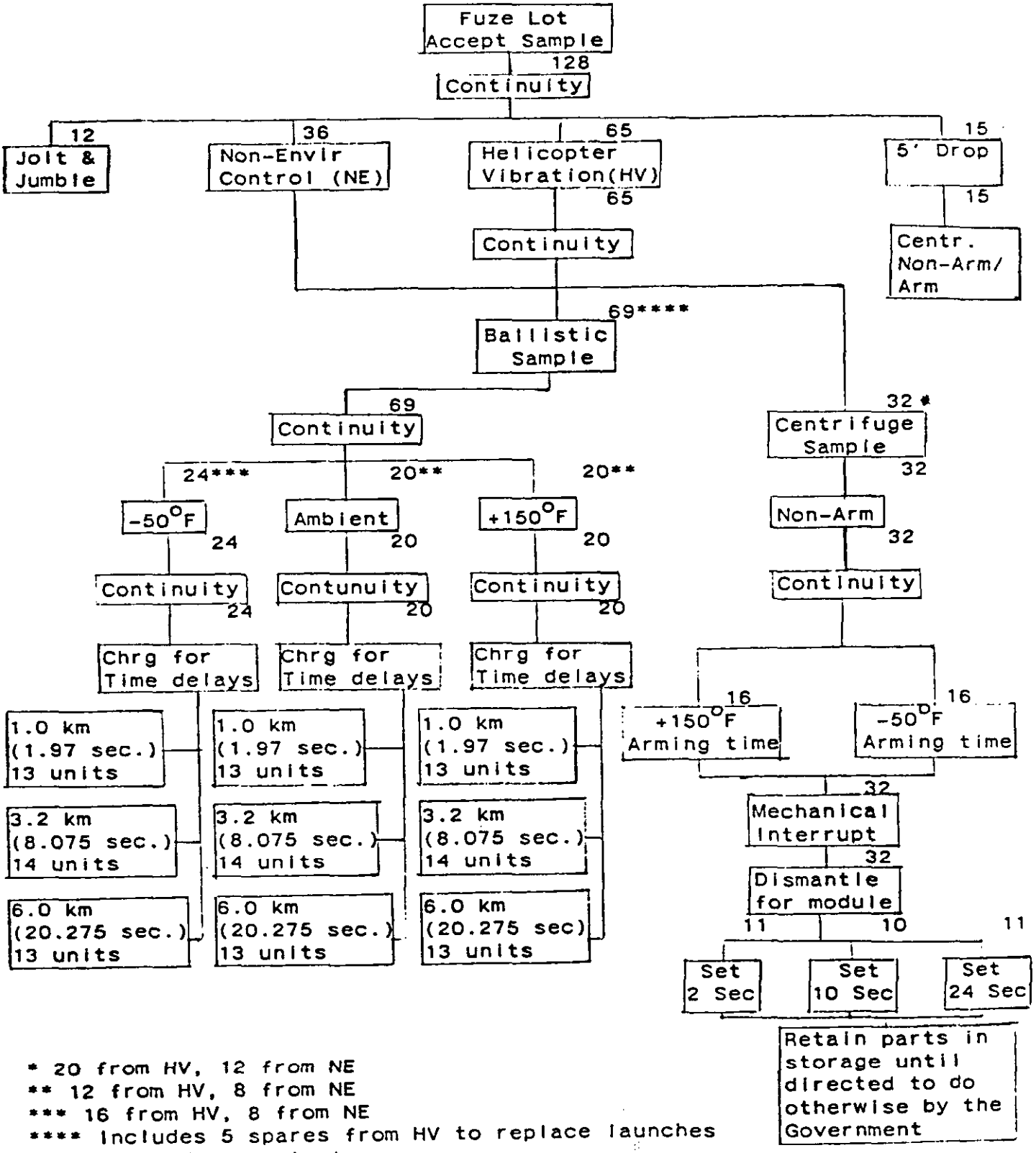
1. All modules shall be serialized.

2. A setter that represents the helicopter setter system shall be used for acceptance of all modules.

Nominal time settings	Environment	Module Quantity	Acceptance Limits (sec.)
2 sec.	-50°F +70°F +150°F Thermal Shock	50 50 50 12	1.96 to 2.04
10 sec.	-50°F +70°F +150°F Thermal Shock	50 50 50 12	9.80 to 10.20
24 sec.	-50°F +70°F +150°F Thermal Shock	50 50 50 12	23.52 to 24.48

TABLE VI. Module time delay test.

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* 20 from HV, 12 from NE
** 12 from HV, 8 from NE
*** 16 from HV, 8 from NE
**** Includes 5 spares from HV to replace launches scored as no-tests.

Note: A setter that represents the helicopter setter system shall be used for delay time acceptance tests in all cases.

TABLE VII. Fuze lot acceptance test.

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